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An image information fusion based simple diffusion network leveraging the segment anything model for guided attention on thermal images producing colorized pedestrian masks

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ABSTRACT

Because of the synergistic characteristics of the data they provide, thermal imagers are becoming more and more common as secondary data-collecting modules to complement classical optical imagers. Their application is especially beneficial when ambient elements like light, smoke, or other particulates need to be handled. As a result, fusion-based detection technology, like assistive driving systems, is trying to incorporate thermal imagers. In this work, we demonstrate that it is possible to create a mask for pedestrians in thermal imaging by building a basic Denoising Diffusion Probabilistic Model and using the results from the Segment Anything Model. Moreover, we show how the anticipated mask can fuse RGB and thermal pictures into a 3-channel output. Finally, we demonstrate how the loss function optimizes the end-to-end analysis in this field. Results from both the quantitative and qualitative analyses validate our methodology.

1. Introduction

A deep learning model created by Meta specifically for image segmentation is called the Segment Anything Model (SAM) [1]. SAM can create more accurate object-specific masks that concentrate on specific areas inside an image, or it can function independently to create global masks for every object in the image itself. Despite being a cutting-edge segmentation method, SAM requires cues to be present to analyze unseen data because the output segmentation masks lack contextual information. The lack of ground truth resources for training, like the KAIST Pedestrian Multispectral dataset [2], which offers ground truth knowledge regarding pedestrians in thermal photos, has resulted in a need for more cues for Thermal Infrared (TIR) images.

TIR images capture passive radiation emitted by objects, contrasting with Near Infrared (NIR) or optical RGB images, which rely on reflected light. However, TIR data is inherently noisy and more challenging to process than RGB images due to its higher wavelength and lower resolution. Consequently, objects in TIR images often have indistinct outlines, especially when their temperature profile resembles the background. Nonetheless, TIR images offer the advantage of being unaffected by external factors such as dust and illumination.

We leverage the KAIST Pedestrian Multispectral dataset [2] to create ground truth annotations for precise pedestrian masks in thermal images. However, generating such annotations with SAM is challenging without specific object masks since SAM predicts all segmentations in an image without direct object knowledge, as demonstrated in Fig. 1. We demonstrate the feasibility of training a simple diffusion network with a Mean Squared Error (MSE) loss function using these object-level binary masks to predict colorized pedestrian instances in thermal images through image fusion and machine learning techniques.

While existing CNN-based fused pedestrian detection approaches [24,25,43,45–47] focus on TIR-RGB fused images, they are often domain-specific or rely on both TIR and RGB images simultaneously. We propose an end-to-end image diffusion-based deep learning model that takes only a TIR image as an input and produces a region-focused fused image containing colorized pedestrian instances independent of multi-domain input images or external environmental factors.

Previous works on tight pedestrian mask generation for the KAIST dataset [3,4] are limited by the unavailability of model weights. Although we aimed to compare our results with prior works, attempts to contact the authors were unsuccessful. However, we were able to find a recent work on pedestrian detection in thermal images, which aims to train an object detector like YOLO [25,28] to initialize the base model and then use a new concept called Disentanglement Then Restoration

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(DTR) [42] to learn a content feature consistency for pedestrian detection in low-resolution thermal images. We include an ablation study on different variations of our underlying model in Section 3 and compare our mask generation results on TIR images against MaskRCNN [30] and DTR in Table 3 and Fig. 4. In summary, our key contributions in this work include:

- Demonstrating the feasibility of training a diffusion model on complex TIR data using segmentation masks generated by SAM.
- Establishing MSE as a numerically stable loss function for multi-domain diffusion tasks, including joint segmentation and colorization.
- Creating a segmented 3-channel fusion output for pedestrians incorporating TIR and RGB data for downstream tasks trained to operate jointly on both domains.

2. Related works

In this section, we present an overview of traditional methods for pedestrian detection, followed by an examination of deep learning-based approaches. These methods include CNN-based methods, Generative Adversarial Networks (GANs), transformer architectures, and vision saliency-based methods for pedestrian detection. Additionally, we discuss fusion techniques for combining thermal and RGB images and methods for creating tight masks, as they bear significant relevance to the current study.

2.1. Traditional feature methods

Using traditional features for pedestrian detection involves identifying and leveraging specific visual attributes within images. As highlighted in [18], such approaches typically entail pedestrian classification, consideration of relative positions, or the creation of Regions of Interest (ROIs).

Methods based on handcrafted features, such as the utilization of Contour Saliency Map (CSM) coupled with background removal from thermal images containing pedestrians, represent one of the earliest works in this domain, as demonstrated by Davis et al. [19]. This approach employed the AdaBoost classifier in combination with the output image from CSM to delineate probable pedestrian regions. While this was a fast method to identify humans in thermal images, it failed when the objects of interest were grouped together.

We also observe another approach in [20], where pedestrian detection was done using the maximum body temperature in an ambient atmospheric environment. While this used a handcrafted feature, it was constrained to apply to a specific range of temperatures and was unsuccessful in generalized conditions.

Jeong et al. [21] further built on this methodology using Cascade Random Forest (CaRF) classifier along with low-dimensional Haar-like features, and Oriented Center-Symmetric Local Binary Patterns (OCS-LBPs) for detecting sudden pedestrian crossing in thermal images. This was based on the premise that during summer nights, the difference in temperature for pedestrians and the roads is similar or slightly different.

As we can observe from the above works, handcrafted features for detecting pedestrians in thermal images are constrained to particular use cases. As such, they are usually unsuitable for generalized cases due to their inability to model complex scenarios.

2.2. CNN based methods

Liu et al. designed one of the first thermal and optical feature fusion methods [22] in thermal images, using CNN to focus on regions containing pedestrians. The problem with this method was that we needed to model every case to work manually. This problem with generalization was partially tackled by [23], where the authors use feature fusion.

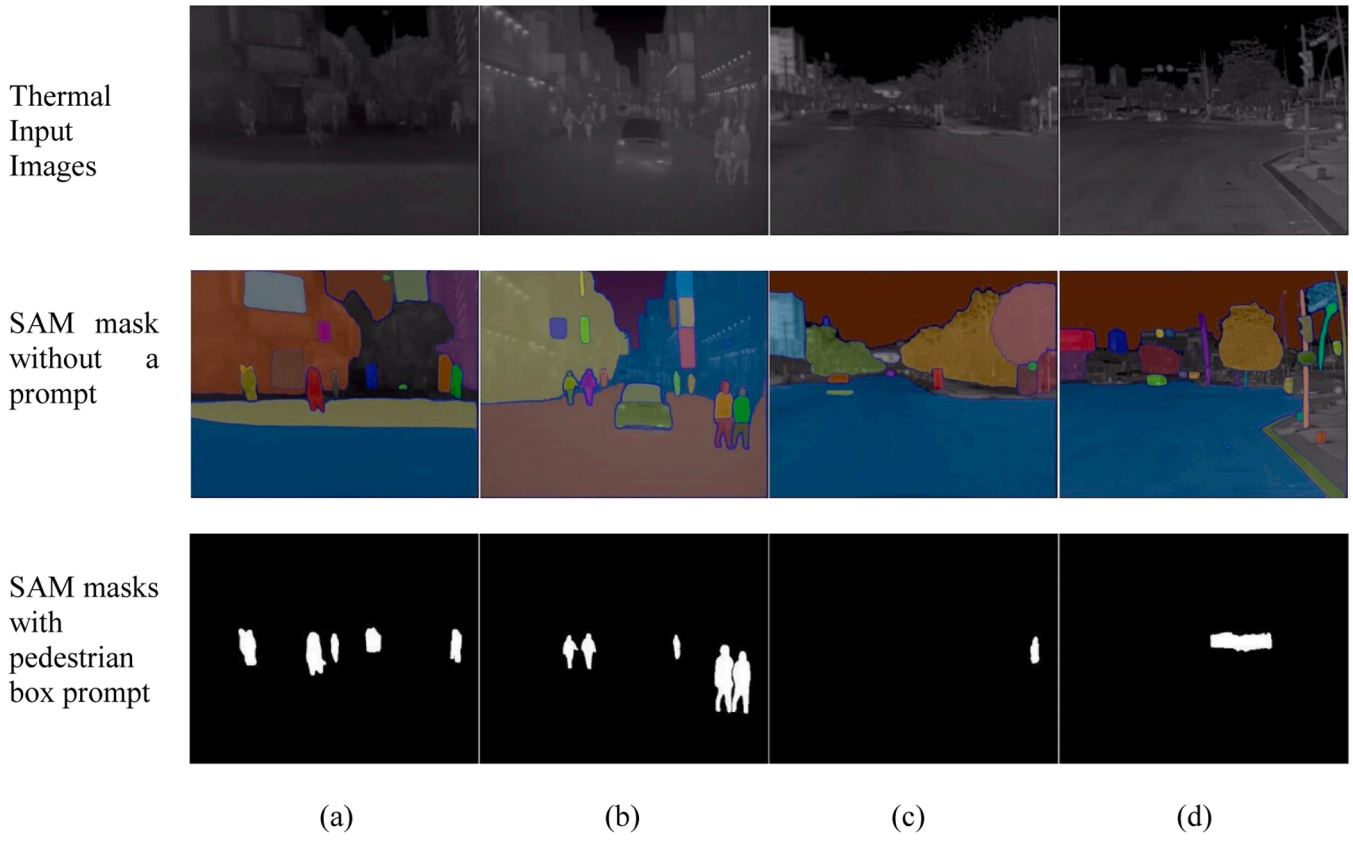


Fig. 1. A comparison of SAM on thermal images with and without prompts for pedestrian mask creation.

In recent years, we have also seen the fusion of thermal-RGB images as a four-channel input image [24] using the You Only Look Once (YOLO) v.3 methods [25] for detecting pedestrians in fused input images. The authors in [26] further improved this by using YOLO v.4, which they named YOLO-4L for small-scale pedestrian detection. Other works like multimodal deep learning-based work similar to the one by Dasgupta et al. [43] and [44–47] rely on multiple sensors to find pedestrians. These methods have a limitation on the availability of data from more than one sensor to work correctly.

Bao et al. [27] used YOLO v.7 [28] to integrate attention fusion and fusion shuffle to help with false detection and further reduce redundant features during detection. Similarly, DTR [42] tries to learn a feature map from visual and up-sampled thermal images to learn a feature map to calculate optimal pedestrian locations in the thermal images.

However, all of the above methods create a box for identifying the region of pedestrians, as shown in Fig. 4. While this is suitable for fast detection use cases, occlusion and fine-grained detection cannot be satisfied with these methods. Examples include autonomous vehicles, medical image segmentation, robotics, and automation.

This led to methods like U-Net [29] and MaskRCNN [30], which can generate tight segmentation masks. This was demonstrated in [48], where Akshatha et al. demonstrated that a combination of Faster R-CNN and Single-Shot multi-box Detector (SSD) can efficiently create bounding boxes for human detection in aerial thermal images.

2.3. GAN, transformer and vision saliency methods

GAN, Transformer and Vision Saliency Methods The use of Generative Adversarial Networks (GANs) in image fusion was introduced by Ma et al. in [31], aptly titled Fusion-GAN. However, this method suffered from the difficulty of training for fused images. Li et al. tackled this problem [32] using a Wasserstein distance-based GAN. More recently, Thermal-Enhancement GAN (TE-GAN) [32] handled this, where the authors enhanced the contrast, combined with noise elimination and edge restoration, to improve the final output.

Another way to look at this problem is to use the most feature-rich module when multiple data sources are available to create a fusion map [33] followed this method by introducing a Cross Modality Fusion Transformer (CFT). CFT handled the fusion by looking at extended dependencies and global context. Specifically, CFT uses the transformers' self-attention mechanisms to handle inter and intra-modal interactions to create a more robust method.

MultiSpectral Pedestrian DEtection TRansformer (MS-DETR) [34], on the other hand, focuses on the extraction of two parallel CNN-based backbones focusing on multiscale feature maps for handling the fusion information.

Saliency maps operate on the principle of constructing a map that highlights regions deemed significant based on the intensity of each pixel. This concept is exemplified in studies such as [35], where Wang et al. employ Multimodal Salient Object Detection to pinpoint specific regions containing detected objects. Marnissi et al. propose a bi-spectral image fusion scheme [36], leveraging a Visual Salient Transformer (VST) with YOLO v.3 as the backbone for expedited detection. Additionally, semantic-level fusion of data from infrared and visible images is explored in [37], where a coupled network is devised to generate fused outputs from joint thermal and visible images.

Existing methods in the domain are constrained by the requirement for joint thermal and visible images in the pipeline, resulting in challenges with multimodal data for producing fused output images. To address this, we introduce a network capable of accepting a single thermal image as input and generating a fused output image. Our approach employs image diffusion to generate precise pedestrian masks, integrating them at the pixel level with the output chrominance channels to produce the final fused image. Furthermore, we utilize the Segment Anything Model (SAM) to jointly create precise masks from thermal images for training the network.

3. Proposed method

3.1. Discussion

Cao et al. [5] propose an information attention fusion network to generate a tight mask. However, there are 3 major differences. The first is that our work can be applied to Thermal Infrared (TIR) images instead of multispectral (TIR + RGB) data, making the module more independent and eliminating the need for registered multi spectral datasets. Secondly, a tighter mask ensures that the output is more focused, which is an essential area of research in computer vision [6–8]. Lastly, our output images can be used in downstream tasks since we aim to create a colorized thermal pedestrian mask following the method outlined in [9]. This process makes it possible to use our method on 3-channel RGB images, which makes them more versatile. We outline the pipeline in Fig. 2.

We demonstrate our results on the pedestrian-masked images from the KAIST dataset [2] and an independent dataset [10] containing thermal images. This is important because the objects in [10] look very different from objects in the database we use for training. We also focus specifically on cases where there are pedestrians because the dataset presented in [10] is a general dataset containing objects from 5 different categories, out of which humans form a small subset. We cover both positive and negative results in Fig. 6 and discuss where our method fails in Section 4.

To our knowledge, this is the only work that deals with operating on real-world thermal images to generate fusion-based contextual masks via a diffusion method. Moreover, we leverage the outputs obtained from the Segment Anything Model to parse our deep learning model similar to transfer learning. Given a prompt, we reason that SAM forms a foundational model trained widely on general optical images and works very well. However, generating a prompt for use with SAM is a vast area of research. With this work's help, creating a non-prompted mask in a diverse area, like hyperspectral images, can be widely adopted via a fine-tuned diffusion network. We found one more work that uses thermal images to create pseudo labels for thermal images using SAM [11]. However, that work utilizes synthetic data to generate their output. This makes the data much less noisy, and the dataset is composed of very high-resolution synthetic thermal images. This is not feasible as such thermal imagers' availability is difficult, compared to the data we use. Moreover, synthetic thermal data is complex to model as well.

The Segment Anything Model uses a loss function that is a linear combination of the Dice Loss [11] and the Focal Loss [12], similar to [13,13]. However, since we already used SAM as the base for our method, we wanted to see if it was possible to simplify the deep learning computer vision model to leverage the already generated tight masks by using a simpler loss function for the fusion task.

We calculate Dice loss according to Eq. (1).

$$D = 1 - \frac{2\,|X \cap Y| + \varepsilon}{|X| + |Y| + \varepsilon} \tag{1}$$

In Eq. (1), X and Y denote the 2 set of contours, the prediction and the ground truth, and ε acts as a small value which prevents the zero division when both X and Y are empty. Dice loss is used when there is a lot of class imbalance, and false positives (classifying a background pixel as a target pixel) and false negatives (classifying a target pixel as a background pixel) need to be equally penalized. If we compare this to the Cross-Entropy (CE) loss where a binary mask is being generated with the pixel wise error thresholded to 0.5, they become equivalent.

On the other hand, the Focal loss is calculated as Eq. (2).

$$F = -\sum_{i=1}^{n} \alpha_i\,(1 - X_i)^{\gamma_i} log\;\; X_i \tag{2}$$

In Eq. (2), X_i denotes the probability of the predicted class membership, α_i is the weighing factor for a given class, and γ_i is the focusing

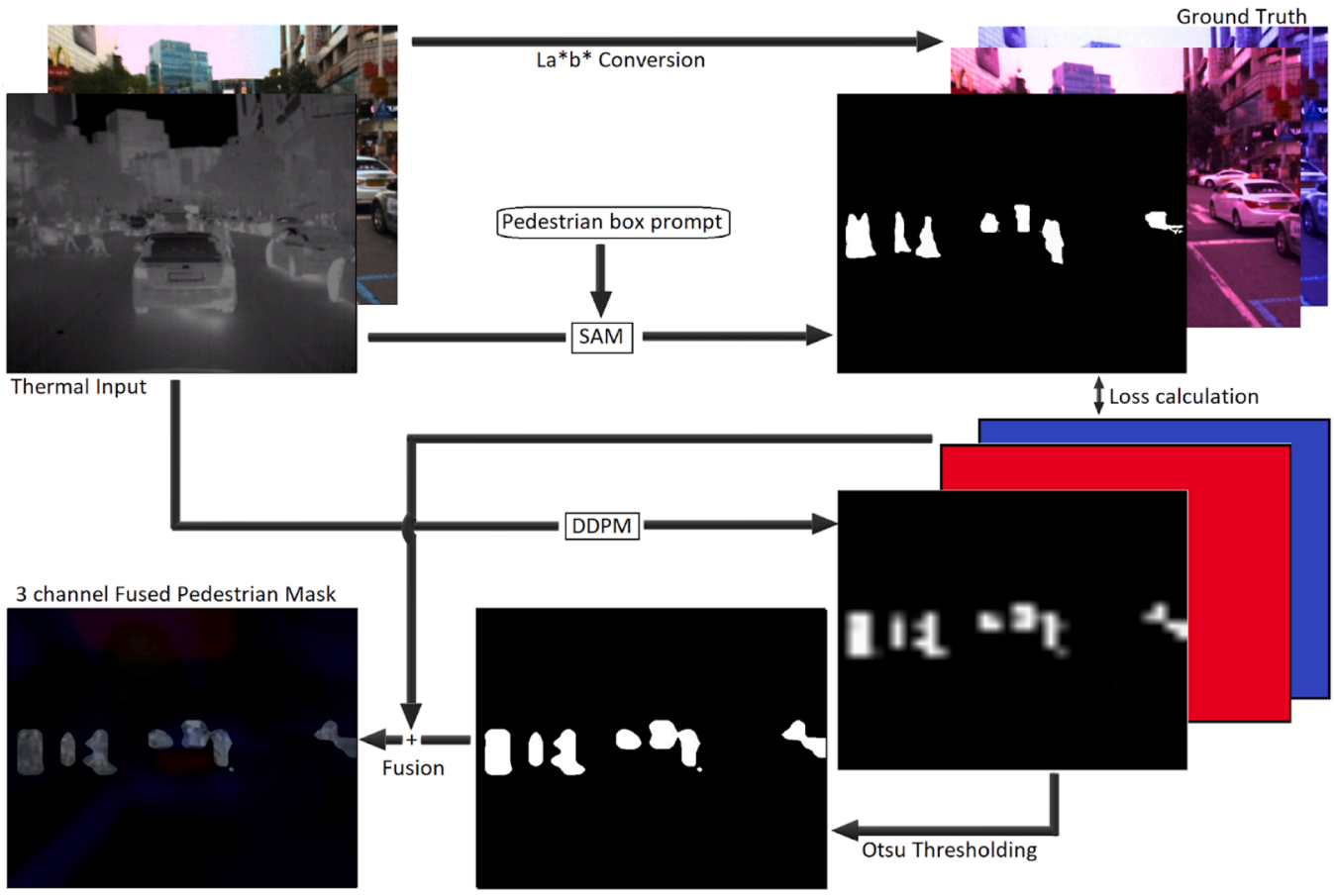


Fig. 2. Method pipeline.

factor. When we have a large value of γ, the focus on a single class increases. However, if we make $\gamma = 0$, the loss becomes equivalent to Cross Entropy loss. Hence, we can see from Eqs. (1) and (2) that both losses in SAM can be reduced to the Cross Entropy loss if we focus on a binary image, which is our specific case for segmentation.

Now, the Cross Entropy loss can be represented as Eq. (3).

$$CE = -\sum_{i=1}^{n} Y_i X_i \tag{3}$$

where Yi denotes the ground truth and Xi denotes the class probability. However, Cross Entropy works only in the case of a mask creation. We are further calculating the mask and the normalized chrominance values of a given thermal image. We assume that our data is Gaussian in nature, wherein a Gaussian distribution of data xi, given its mean (μ) and its standard deviation (σ) can be represented as Eq. (4).

$$p\ (x_i,\ \mu,\ \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\frac{1}{2}\frac{(x_i-\mu)^2}{\sigma^2}\right) \tag{4}$$

We assume that our data is Gaussian since the base work on Denoising Diffusion Probabilistic Model [14] assumes the same. Now, the optimization function for any deep learning method can be structured as a function of maximum (or minimum) likelihood. As such, we can restructure our loss as the log-likelihood of a probability function between the ground truth (X) and the prediction (Y) so that the domain changes from multiplicative to additive functions to counter the underflow cases. This can be written as Eq. (5).

$$f = \sum_{i=1}^{n} p(Y_i \mid X_i;\ \theta) \tag{5}$$

In Eq. (5), θ represents the parameters of the given loss function. Thus, if we apply the natural logarithm on Eq. (4) and sum over the obtained expression, our expression reduces to Eq. (6).

$$\begin{aligned} log(p) \\ &= \Sigma\ log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \Sigma\left(-\frac{1}{2}\frac{(x_i-\mu)^2}{\sigma^2}\right) \\ &= log\ 1 - log\left(\sqrt{2\pi\sigma^2}\right) - \frac{1}{2}\Sigma\frac{||x_i-\mu||^2}{\sigma^2} \\ &= K - \Sigma\frac{||x_i-\mu||^2}{c} \end{aligned} \tag{6}$$

In Eq. (6), K and c are constants. This is because, if we train our method on a particular distribution of data (in our case, the KAIST Multispectral pedestrian detection dataset [2]), the value of σ becomes a constant dependent on the whole data. However, the mean (μ) varies based on the particular data point, similar to the work in [14]. Hence, our method reduces to a function of the ground truth (X) versus a function of the predicted data value (Y). We can combine Eqs. (5) and (6) and rewrite this as Eq. (7).

$$f\ (X, Y) = K - \Sigma\frac{||\ X_i - Y_i\ ||^2}{c} \tag{7}$$

Thus, Eq. (7) reduces to a form of the MSE loss function. We reduce our loss function to MSE because Cross Entropy loss cannot be used for calculation of normalized chromatic values (shown in Fig. 5), used for the fusion output. While MSE is unbounded in both upper and lower direction, we are using a normalized image output ground truth, passed via sigmoid functions at each upscaling and down scaling layer. Given a data value of x, the sigmoid function can be written as Eq. (8).

$$S = \frac{1}{1 + e^{-x}} \tag{8}$$

This practically lets us experimentally bind our MSE loss function to a range of [0, 1]. We show that this is a valid approximation in our qualitative results in Fig. 5. This proves that MSE is a valid loss function when we want to train a Denoising Diffusion Probabilistic Model (DDPM) [14] network for working on complex data, related to segmentation. We also create a variation of the mask network alone (without the chromatic layers), trained on CE loss function. We show the same scores in Table 3 and the mask image outputs in Fig. 4.

We add the Gaussian noise according to the noise schedule in [14] as Eq. (9).

$$q(X_t \mid X_{t-1}) = N\left(X_t;\ \sqrt{1-\beta_t}\ X_{t-1},\ \beta_t I\right) \tag{9}$$

In Eq. (9), the distribution for creating an image $X_0 \sim q(X_0)$ refers to the output of the forward diffusion process. This method adds Gaussian noise at each time step t, given a variance schedule that is known a priori as $0 < 1 < 2 < \ldots < t < 1$. Here, β_t refers to the forward process variance, which can be learned by reparameterization, as denoted in [41] or simply be set to a constant. It needs to be noted that if we set β_t to a small value, both the forward and the reverse process for diffusion hold the same functional form, simplifying the complexity of the operation. In our case, we set $t = 1000$. We refer to the work at [14] for this.

Image diffusion is a process within the image processing domain wherein an image undergoes spatial smoothing through the redistribution of pixel values across the neighbouring regions. This diffusion mechanism reduces image noise while preserving significant features by iteratively adjusting pixel intensities based on local characteristics. We use this characteristic of image diffusion to focus on localized regions containing pedestrians in thermal images to create a thermal-RGB fused pedestrian mask.

Various diffusion algorithms, such as the popular Persona-Malik model or the more recent deep learning-based diffusion [14], operate by iteratively diffusing pixel information according to predefined rules, typically involving gradients or statistical measures of similarity. This iterative diffusion attenuates high-frequency noise, resulting in a smoother image while maintaining essential structural information. Since we are dealing with binary masks, this property of diffusion methods helps us iteratively define a smoother mask in our case.

Our database comprises 45415 images, all the thermal-optical pairs annotated to show pedestrian locations in the images. The process that we follow is shown in Fig. 2. The full model comprises 2 submodules. The first is the SAM, used with the box annotations from the KAIST dataset [2] to create the tight binary masks that we use as a section of the ground truth for the diffusion model we train. The second module is the diffusion model we train using the DDPM, with U-Net as the backend.

3.2. Data preparation

There are 2 levels of data we use for our study:

- Creation of masks from SAM
- Creation of fused thermal-RGB data

The KAIST dataset [2] has pairs of thermal and optical images for both day and night time. These images come with boxes denoting pedestrians. We use these boxes as an input for the thermal images for the Segment Anything Model. Without these box prompts, SAM segments the full image, as shown in Fig. 1. However, if we provide a box prompt, SAM can localize the instances of the pedestrians accurately and provide a tight mask for the pedestrians. These masks form a part of the output image, and is also the pedestrian localization mask. We fuse this mask with the input thermal image to form the luminance channel of our fused output.

On a parallel branch, we take the corresponding RGB image for the thermal input, and obtain the chromatic information of this channel via La^*b^* conversion of the RGB image.

Once we have the mask from SAM, we use it with the thermal image to form the masked thermal image, which we use as the output luminance channel. We combine this along with the 2-channel chromatic information to create a 3-channel data of the same size as the input image, which we finally transform back to the RGB domain. We use this as the ground truth for the DDPM and train against the thermal image as an input to create our final model. Furthermore, we have used one step in post-processing the generated mask via Otsu's thresholding, which we explain in Eq. (10). We show the complete pipeline for the data processing in Fig. 3.

It needs to be noted that the RGB image is only used during the pipeline training. The test case only requires a thermal image as an input, and the pipeline is built to provide without the use of a registered RGB image, since this information might not be available in real world scenarios.

3.3. Ablation study

We show the results we obtain from 3 different variants of the network to show how the results differ with various architecture changes made to the DDPM model for the mask generation. For the base case, we fine tune the model available at [15], which is an implementation of [14] to show how the results look on a base U-Net based diffusion computer vision model. We use the checkpoint provided by the authors for the training of this model. We propose three different Models with varying levels of encoders and Intermediate layers to show how the results vary. These 3 models are trained from scratch, without any backbone, so that there is no bias in the output. These results are presented in Fig. 4; the quantitative results are available in Table 3.

We implement the base U-Net model with a backbone trained on ImageNet. The basis of diffusion is an input perturbation based on time step, which effectively makes the perturbation random, and adding this variation at each step of the model in the form of noise as indicated in Eq. (9). At the final output, this random noise is predicted and subtracted from the input variable (in this case, a fusion based image) to create the final output. We simply remove the last step of the method, and model the diffusion network to generate a binary mask instead of a random noise. Since the perturbation is still random, the method effectively learns to create the mask from random data, which looks similar to the input thermal image data, but with some noise added in. This model uses an encoder with a final patch size of an eighth of the input size and an intermediate layer consisting of 6 convolution layers at the lowest patch size.

For the second variation, we simply take the images down to a 2×2 sized encoder (which is 1/16th of the input image size) and then upscale it back to the full size. At 1×1 level encoders, the network focuses on pixel level information. However, since we need the object level masks for pedestrians, we constrain our feature map to 4×4 patches [16]. We preserve skip-connections and intermediate layers between the encoder and decoder blocks of the same size, like the original U-Net implementation, to ensure that the information is not lost while encoding. The skip connections also work to preserve the feature maps at different encoder resolutions to indirectly help with pedestrian depth in the thermal images.

The third variation, which is the mask generation model we are proposing, uses encoders only till 4×4. Considering that our input size is 32×32, this means an effective encoding up to an eighth of the original input size. We chose this variant as our output method because a statistical analysis of the data showed that the mask size for the thermal images in the KAIST dataset [2], on average, is one-eighth of the input size. We also remove the intermediate layers, and connect the output of the encoder to the input of the decoder directly. This was done because at the lowest patch level, these layers would be similar to the addition of noise in a binary image. The self-attention layer already adds noise to

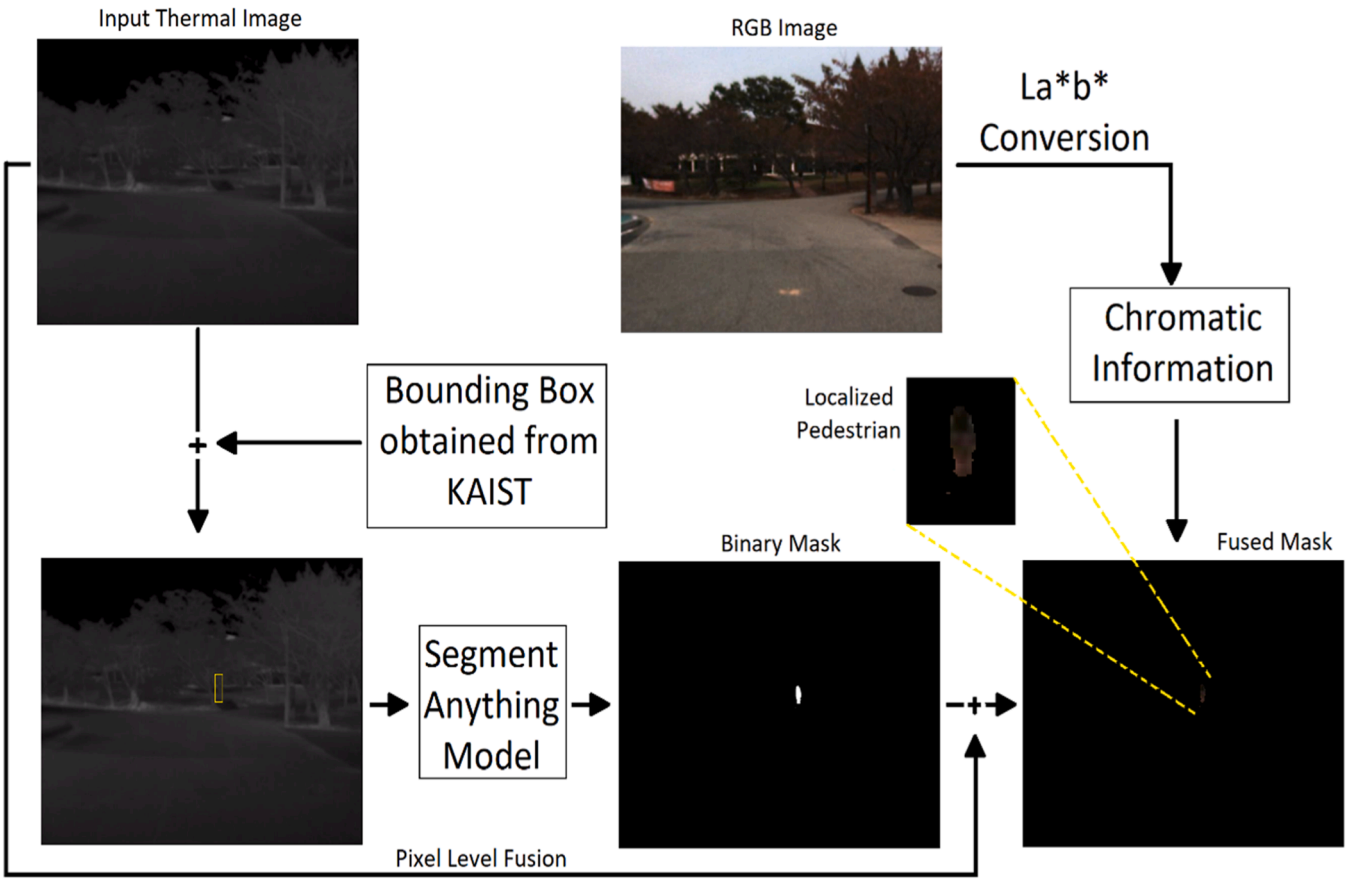


Fig. 3. Data preparation pipeline.

the model in a controlled way using Eq. (9), and this extra noise would likely have made the model more unstable. This hypothesis is also backed by the quantitative scores we observe in Table 3.

Finally, we retrain the proposed method to output a 3-channel array. The first channel of this is the binary mask, normalized to [0,1] range, applied on the thermal image to form a localized tight pedestrian masked thermal image. The second and the third channels are the 2 chrominance channels obtained from the La*b* image of the RGB ground truth. We normalize these two channels to conform to the [0,1] range by adding 128 to the chromatic channels and dividing it by 255. This is so that the output fits the range of the sigmoid activation functions of our deep learning model for each upscaling and downscaling operation for fusion. We use the image processing technique of Otsu's thresholding, as shown in Eq. (10), on the final generated mask ($m_{x,y}$, given a data point x,y) to make it binary for better output segmentation.

$$m_{x,y} = \{0 \text{ if } d_{x,y} < \textit{threshold } 255 \textit{ otherwise} \tag{10}$$

The lower bound in Eq. (10) is 0 as our image is binarized to [0, 255] range before we process it. We set the threshold to 127 as the midway value between the 2 values constituting the binary image. All the results from different methods are shown in Fig. 4.

3.4. Deep learning architecture discussion

We train our network to work on 32 × 32-sized images for 2 reasons. DDPMs are computationally very expensive, and a size of 32 × 32 keeps the network lightweight for deployment on a wide range of solutions. The second reason is that we are calculating a combination of a binary and a chromatic mask, which is easy to upscale. If the data were complex RGB images, this would result in artifacts and blurriness, which is undesirable. For a binary mask, this is no longer a problem since there are only 2 values, which a simple linear interpolation is able to easily handle.

However, for the mask generation networks, we process the input thermal images as well as the output binary masks as 3 channel images where all 3 channels have the same information, in order to keep it comparable to the original work, which is a 3-channel image-based work. This also makes the final proposed method a transfer learning process using the trained model we obtain from the 3-channel mask output method using MSE. This reduces the number of training steps needed since the initial channels need minimal retraining, and the maximum training happens only in the last 2 layers. However, the benefit of using a 3-channel input is that the backbone can be reused for downstream tasks directly for transfer learning on 3 channel fusion based thermal-RGB images. Finally, we post-process the output mask using Otsu's thresholding explained in Eq. (10) to make it a true binary image and fuse it with the thermal image and the chromatic output to generate the final output image. In more detail, we explain the model architecture in Table 1 and make the code available at [17].

In Table 1, each of the *down* blocks downsample the data from a size of (h x w) to (h/2 x w/2) using a Max Pool layer. The *up* blocks use a Bilinear upsample layer to do the reverse. All *down* and *up* blocks also have a Self-Attention layer at each level, equal to the output layer depth so that the method learns the attention details for the fusion based diffusion process. Detailed information on each of the layers is explained in Table 2.

In Table 2, *x_1* refers to the output from the Encoder block at the same resolution, that is concatenated to the output of the decoder layer. *GeLU* refers to the Gaussian Error Linear Units activation function. This is represented as Eq. (11).

$$GeLU(x) = x * \Phi(x) \tag{11}$$

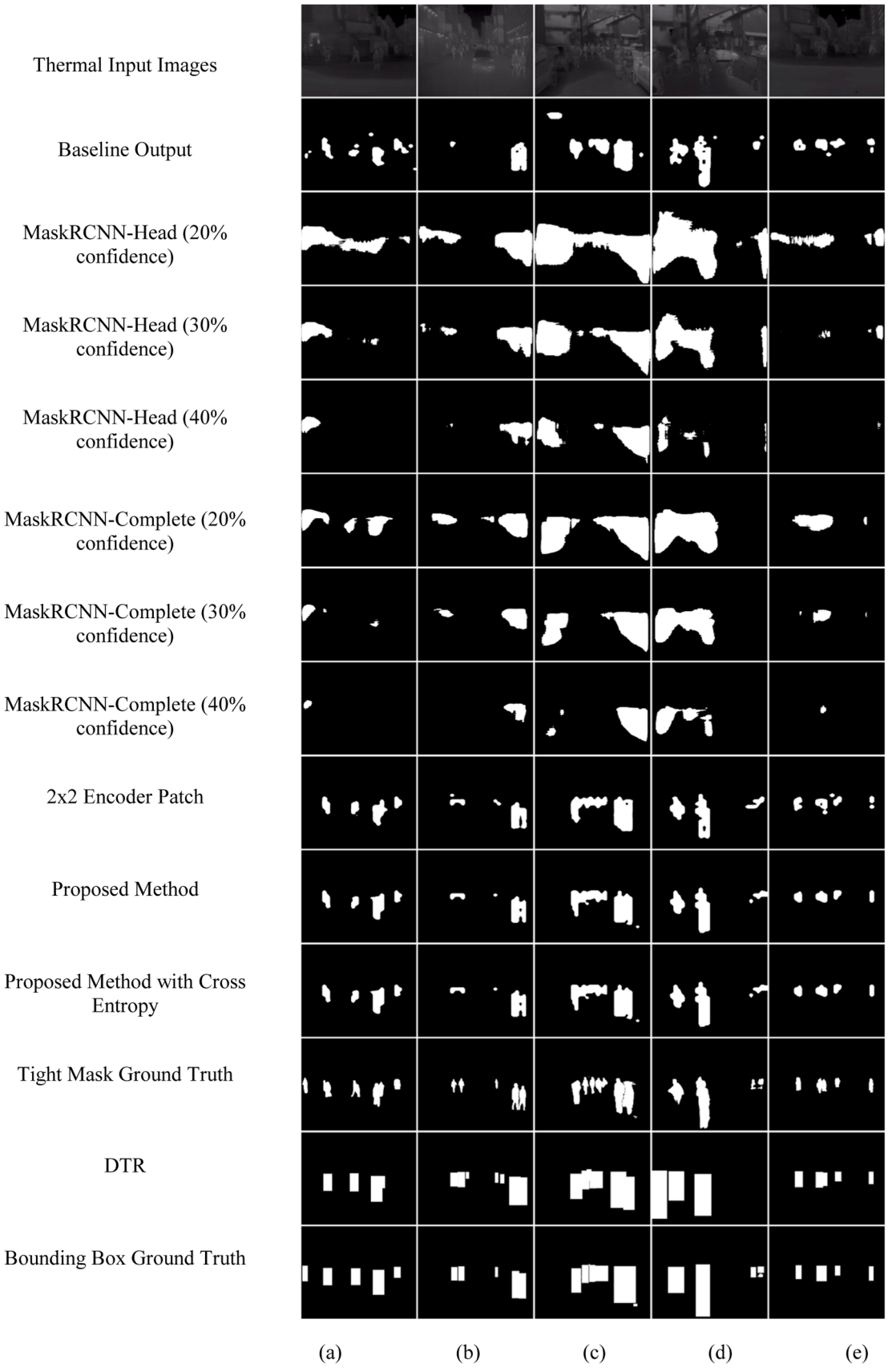


Fig. 4. Comparison of masks produced by different methods.

Table 1
Deep learning network description.

Block 1: Encoder	Block 2: Decoder
doubleConv (3, 64, 64)	up (512, 128)
down (64, 128)	up (256, 64)
down (128, 256)	up (128, 64)
down (256, 256)	doubleConv (64, 3, 3)

In Eq. (11), $\Phi(x)$ is the cumulative distribution function for the Gaussian Distribution, given an input distribution x. We use *GeLU* as an activation function in our method because it has been shown to prevent overfitting in DDPM since in comparison to other activation functions like the Rectified Linear Unit (*ReLU*), *GeLU* works better when the neurons have negative values associated with them. This is because, in comparison to *ReLU* and *Elu*, which are convex and possess monotonic activations, the increased curvature and non-mono tonicity of *GeLU* can allow for more complex approximations [40].

We use the base Denoising Diffusion Probabilistic Model (DDPM) architecture as the baseline for all of our experiments. DDPM is based on the encoder-decoder architecture, but with the use of embedded Self-Attention for helping the network learn where to focus on a thermal image. As shown in [14], we use a timestep=1000 with a U-Net style backbone for our experiments. The Upsample layers use bilinear upsampling, and the parameters are shared across the time domain using Transformer sinusoidal positional embedding [38] in both the Up and Down layers of the network, as indicated in Table 2.

Following the method shown in [14], we use self-attention at the 16 $\times$ 16 feature resolution and in the subsequent lower encoder levels. These are concatenated with the encoder blocks at the same feature resolutions to ensure that the local encoder information is not lost across the layers. Finally, we use a convolution layer with a 1 $\times$ 1 filter and the necessary output channels to obtain a pixel level convolution output.

The complete code and trained weights are available at [17] for further reference. We also provide the complete data created as a part of this work at [39] for reproducibility.

4. Results and discussion

Our method aims to create a 3-channel image information fusion output mask, containing RGB and thermal information in the output image. This ensures that any task that needs a 3-channel output can leverage the color information from the chromatic channels, while retaining the thermal information from the luminance channel. As such, we fuse the mask with the thermal image at pixel level using the equation outlined in Eq. (12).

$$output_{x,y} = thermal_{x,y} * \left(mask_{x,y} \;/\; 255\right) \tag{12}$$

In Eq. (12), we normalize the deep learning output mask to a [0, 1] range to get a binary output for the thermal image, which we can use as the luminance channel for our image information fusion 3 channel La*b* image. This is a standard machine-learning technique generally used for image processing. We finally convert the La*b* image to RGB for the output image. We show our output in FigS. 4 and 5.

Likewise, we use the code for [14] as the baseline for our experiment because this work establishes the method to implement the Denoising Diffusion Probabilistic Model. The difference between the baseline method and the proposed one (along with the methods used for the ablation study) is the absence of the intermediate convolution layers, which work on the lowest feature maps of the encoder block. We show that this is optional in our use case because the 4 $\times$ 4 patch size is lower than the smallest mask present in our dataset. This ensures that the feature map at this level is already highly optimized, and our hypothesis was that convolving further on these windows might degrade the output instead of making it better. This can be validated with the help of the scores present in Table 3, which denote that our method yields better results.

We show the various mask outputs we obtain with the different variations of our network in Fig. 4. We also compare our results with MaskRCNN [30] trained from scratch with our data along with DTR [42], for which we use the model weights provided by the authors. However, unlike our work, DTR predicts bounding boxes for pedestrians. As such, in order to have a proper comparison, we also provide the bounding boxes for our tight masks and compare the results accordingly in Fig. 4 and Table 3. We use the PyTorch implementation of MaskRCNN along with the ResNet-50-FPN backbone for our experiments.

We use 2 variations of MaskRCNN for our comparisons: one where we freeze the network layers except for the last layer (head) with the weights from ResNet50-fpn and the other where we retrain the entire network from scratch. Further, we use 3 levels of confidence scores (20 %, 30 % and 40 %) with the network, in order to obtain the mask outputs. We plot all these output images in Fig. 4.

We can observe that MaskRCNN cannot properly segment the pedestrian masks in Fig. 4. In both the completely retrained and also the partially trained (Head retrained) variants, we see that MaskRCNN either overestimates the mask region containing the pedestrians (Fig. 4 (b), 4(c), 4(d)) or completely misses sections of the pedestrian mask (Fig. 4(e)). This trend is observable across all confidence levels of MaskRCNN. The confidence level determines the confidence of the network in the predicted masks. The network completely fails to predict masks at higher confidences, so we do not plot results above a 40 % confidence.

In the case of DTR, we do not retrain the model with our data; instead, we use the pre-trained model. However, we observe that the method is unable to segment the pedestrians in around 25 % of the test cases. This results in a complete black mask on the output domain, reflected in the scores we observe in Table 3, where the SSIM score for the method is the lowest of all compared methods. Since DTR calculates bounding boxes for pedestrians in thermal images, we also provide the Ground Truth bounding boxes for our tight masks in Fig. 4.

We can observe that in Fig. 4(b) and 4(d), the method is computing an erroneous pedestrian in the middle and at the left-hand side of the image, whereas, in 4(a), it is unable to find the smaller pedestrian at the

Table 2
Layer Description for network in Table 1.

doubleConv (in_channels, mid_channels, out_channels)		up (in_channels, out_channels)		down (in_channels, out_channels)	
Layer Type	**Parameters**	**Layer Type**	**Parameters**	**Layer Type**	**Parameters**
Conv2D	kernel = 3, padding = 1, bias = False	*UpSample*	scale = 2, mode = Bilinear	*MaxPool*	-
GroupNorm	1, mid_channels	*concatenate*	-	*GeLU*	x + doubleConv
GeLU	x	*GeLU*	x + doubleConv	*doubleConv*	in_channels, in_channels/2, out_channels
Conv2D	kernel = 3, padding = 1, bias = False	*doubleConv*	in_channels, out_channels, out_channels	*Sigmoid*	-
GroupNorm	1, out_channels	*doubleConv*	in_channels, in_channels/2, out_channels	*Linear*	256, out_channels
		Sigmoid	-	*Reshape*	t
		Linear	256, out_channels	*Concatenation*	x, t
		Reshape	t		
		Concatenation	x, t		

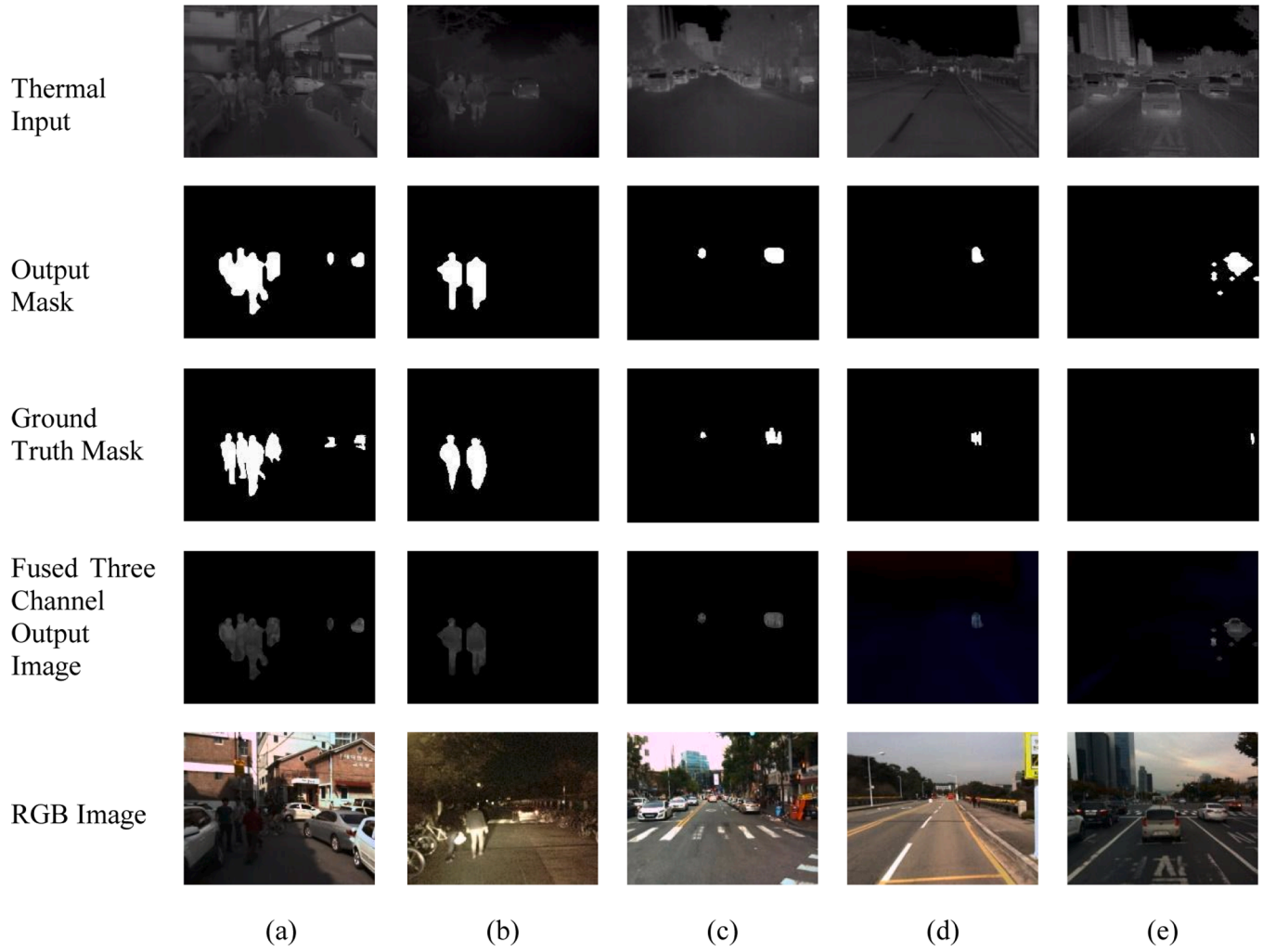


Fig. 5. Results for the proposed method.

Table 3
Quantitative Scores for the masks.

Method	MSE	IoU	SSIM	Dice Score
Baseline	0.063	0.321	0.976	0.446
MaskRCNN-Head (20 % confidence)	0.084	0.087	0.92	0.1679
MaskRCNN-Head (30 % confidence)	0.064	0.1	0.944	0.182
MaskRCNN-Head (40 % confidence)	0.05	0.101	0.961	0.176
MaskRCNN-Complete (20 % confidence)	0.074	0.098	0.933	0.183
MaskRCNN-Complete (30 % confidence)	0.058	0.109	0.952	0.194
MaskRCNN-Complete (40 % confidence)	0.047	0.108	0.965	0.183
2 × 2 Encoder Patch	0.045	0.50	0.952	0.631
Proposed Method with MSE	**0.041**	**0.533**	0.962	**0.653**
Proposed Method with Cross Entropy	0.043	0.53	**0.986**	0.597
DTR [42]	0.054	0.525	0.541	0.465

left-hand corner. This shows that the method detects both false positives and false negatives, resulting in the scores we observe in Table 3.

The 2 × 2 encoder patch refers to the lowest level of encoder, going down to 2 × 2 (instead of 4 × 4 for the proposed method). For all the network variations we propose, the predicted masks for pedestrians are much more precise than MaskRCNN. This result is evident in Fig. 4(d) and Fig. 4(e), where the pedestrian masks are very small since the pedestrians in the image are far from the camera. However, it also needs to be noted that in case of 4(a), the methods all fail to identify the pedestrian at the left edge of the image. This is one of the cases where the baseline method performs slightly better than the proposed one. However, such cases are very rare, and can be alleviated with the help of relevant data for training. This is also reflected by the quantitative analysis presented in Table 3, where all scores are the highest for one of the 2 proposed methods.

The method fails in some cases, especially if the pedestrian's distance is very high, wherein the objects become very small and overlap with complex objects like cars. We show an example of this in Fig. 5(e). This image shows that the ground truth image is a tiny mask of a pedestrian, which overlaps with a car. The pedestrian is very far away from the camera and partially occluded, causing this problem.

However, the problem is less prominent if the pedestrian is not occluded and far away at the same time. We can see this in Fig. 5(c), where the pedestrian in the middle of the image is very far away, causing the mask to be tiny. Similarly, the pedestrians in Fig. 5(d) are partially occluded but have a more extensive combined detection area, causing the proper mask generation. However, the masks are incorrect, causing a slight deviation in the separation between the pedestrians. This can be seen in Fig. 5(a), 5(c) and 5(d). This is partly caused by the fact that we down-sample the image to a resolution of 32 × 32 before feeding it into the diffusion network, and then up-sample the masks back to their native resolution of 640 × 520. This causes the masks to have a smoother gradient than the ground truth. We provide a comparison of the Mean Square Error (MSE), Intersection over Union (IoU), Structural Similarity Index Measure (SSIM), and Dice Score [11] values of the output binary mask images and their ground truth images in Table 3 for all of our methods.

While the scores for the proposed model with MSE loss are similar to those for the proposed model with Cross Entropy loss in Table 3, there are about 10 % output masks that are blank, generated from Cross Entropy loss. This means that Cross Entropy loss rewards proper masks more (as demonstrated by our two highest scores with this method) and equally penalizes bad masks more. However, since we are creating a 3-channel image information fusion method that can output both the binary mask as well as the chrominance data, the proposed method uses MSE as a loss function. Moreover, the proposed method also reports the lowest error and the highest Intersection over Union (IoU) score. It is only slightly worse than the Cross Entropy loss method in SSIM and Dice Score.

We chose a final patch size of 4 × 4 rather than going down to the pixel level because we are focusing on a segmentation mask, and a pixel level local feature would not contribute any significant information on this. Thus, we chose to statistically see the average mask size in our dataset, which was one of the hyper-parameters we tuned in the model.

In addition to the previous results, we also compare our method on an independent dataset of thermal images [10], which we had collected as prior research work. The dataset contains 1873 pairs of (RGB-thermal) images captured via 2 separate thermal imagers and are of widely different resolutions and subjects. There are 5 different categories of images, including buildings, greenery etc. Hence, we filter out images containing humans in the images and run our method on them to show that the developed method works on images of very different types than what the network is trained on. We show these results in Fig. 6.

The dataset from which the input photos in Fig. 6 are taken is not utilized for segmentation. As a result, we are unable to compare ground truth segmentation masks. Nevertheless, the generated masks from Fig. 6(b) show that the model struggles to identify people close to the camera. The reason for this is most likely due to the fact that every one of the training images for the method features pedestrians on the road, all of them at a similar distance to the imager. As such, when persons are close together, the approach cannot segment them correctly in the figure. Due to insufficient similar images, we refrain from training the fusion dependent 3 channel model and instead present our findings on tight mask.

Another essential thing to note is that the background of these images, especially Fig. 6(a) and 6(b), are not roads and are indoor images. In comparison, Fig. 6(c) and 6(d) are outdoor images, and provide a much better output mask, most likely because they are semantically similar to the images used for the training. We can fix this problem by adding more indoor images with human segmentation. However, such datasets are rare, most likely because they take time and effort to be created. Fig. 6(a) and 6(b) are from FLIR E40, with a native resolution of 240 × 320 while Fig. 6(c) and 6(d) are acquired using Sonel KT400, with a native resolution of 384 × 288. We resize these to the model input shape of 32 × 32 before obtaining the output, which we resize back to their default resolution. The experiments were conducted at a resolution of 32 × 32, with a training size of 43144 random pairs of thermal images and their corresponding tight thermal masks and the chrominance information from the corresponding RGB images for fusion. We use 961 separate images for testing. The training was done for around 650 epochs before the loss converged. It took around 284 seconds per epoch with a batch size of 64 images, training on an RTX 6000 24 GB GPU using 128 GB RAM.

5. Conclusion

We suggest a pedestrian mask generator built around diffusion to assist with a fusion of multi-domain images. It is trained to produce three-channel pedestrian masks and may be utilized with techniques designed to operate with RGB images. The result creates a unique mask with information from both domains by combining the thermal plus RGB data. Furthermore, we demonstrate that this type of model may be created using SAM data. Even though the 32 × 32 image size we utilized is quite modest, the technique still functions at this resolution. We also demonstrate how it functions with a collection of data that the system has never seen before, which was gathered by two distinct thermal imagers. Additional research along these lines would include

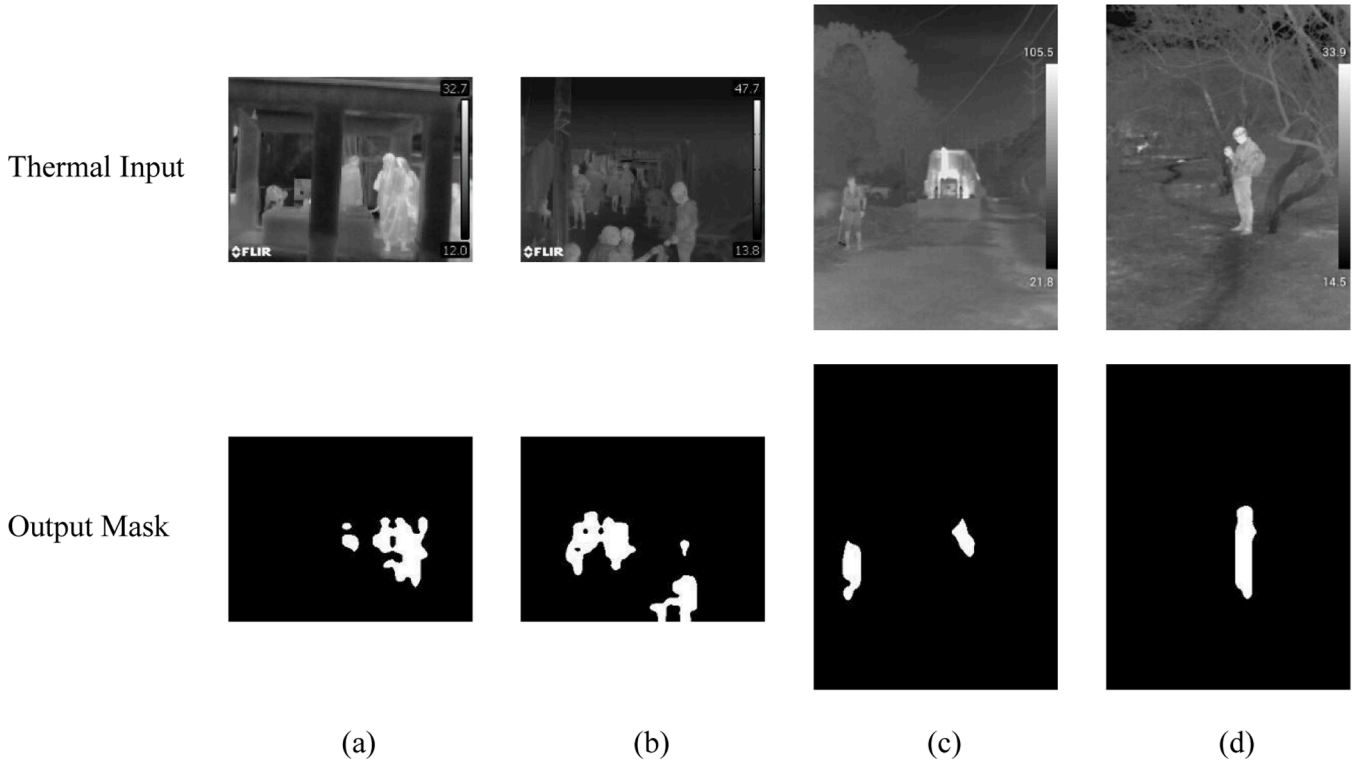


Fig. 6. Output masks on images from the dataset at [10].

experimenting with different loss functions and use a greater resolution to observe how the result improves. Since the existing method has a significant temporal complexity regardless of a low-quality input image, another relevant topic of inquiry would be ways to optimize the technique for translation to applications that operate in real-time.

CRediT authorship contribution statement

Suranjan Goswami: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Satish Kumar Singh:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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